

Analysis PhD Exam, May 2011

Answer SIX questions. Write solutions in a neat and logical fashion, giving complete reasons for all steps and stating carefully any substantial theorems used.

1. This problem has two parts.

(i) State the Hahn–Banach theorem.

(ii) Show that there exists an isometric linear map from each separable normed space into ℓ_∞ .

2. Prove that the spaces c (of all convergent scalar sequences) and c_0 (of all null scalar sequences) are not isometrically isomorphic when equipped with the sup norm.

3. State the Closed Graph Theorem and the Banach Isomorphism Theorem; deduce one of these from the other.

4. Let the sequence $(T_n)_{n=1}^\infty \subset L(X, Y)$ be bounded in operator norm and assume Y complete. Prove that

$$Z = \{z \in X : (T_n z)_{n=1}^\infty \text{ converges}\}$$

is a closed subspace of X .

5. Let X and Y be closed subspaces of a Hilbert space. Prove that if $X \perp Y$, then $X + Y$ is closed.

6. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Define $\overline{\mathcal{F}}$ to comprise all those $A \subseteq \Omega$ for which there exist $L, U \in \mathcal{F}$ such that $L \subseteq A \subseteq U$ and $\mu(U \setminus L) = 0$ and then define $\overline{\mu}(A)$ to be the common value $\mu(L) = \mu(U)$. Show that $(\Omega, \overline{\mathcal{F}}, \overline{\mu})$ is a measure space that is complete in the sense that each subset of a null set is null.

7. Let $(\Omega, \mathcal{F}, \mu)$ be an arbitrary measure space; let $p, q, r > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$. Prove that if $f \in L_p(\mu)$ and $g \in L_q(\mu)$ then $fg \in L_r(\mu)$ with

$$\|fg\|_r \leq \|f\|_p \|g\|_q.$$

8. Let $1 \leq p < q < \infty$. Show that $L_p(\mu) \not\subseteq L_q(\mu)$ iff $(\Omega, \mathcal{F}, \mu)$ contains measurable sets of arbitrarily small positive measure.